Practical No. 10 – One Sample Hypothesis Testing for Population Mean (z-test and t-test)

Blue = given data (inputs). Black = formulas. Each test is solved by two methods: Critical-value and p-value; both must agree.

Ex.1 Burning rate of propellant – z-test (two-tailed)

H0: μ = 50 vs H1: μ ≠ 50 (σ known, so z-test)

Hypothesised mean μ0 (cm/s)	50	
Population SD σ	2	
Sample size n	25	
Sample mean x̄	51.3	
Significance level α	0.05	
Test statistic z	3.2500	$z = (\bar{x} - \mu 0)/(\sigma/\sqrt{n})$
Critical value ±z(α/2)	1.9600	NORMSINV(1−α/2)
p-value (two-tailed)	0.00115	2×(1−Φ(|z|))
Decision (critical value method)	**Reject H0**	
Decision (p-value method)	**Reject H0**	

Conclusion: Strong evidence that the mean burning rate differs from 50 cm/s.

Ex.2 Price of new one-family homes (South) – z-test (left-tailed)

(a) H0: μ = 181900 vs H1: μ < 181900

Hypothesised mean μ0 ($)	1,81,900	
Population SD σ ($)	33,500	
Sample size n	40	
Sample mean x̄ ($)	1,66,400	
Significance level α	0.01	
(b) Test statistic z	**-2.93**	$z = (\bar{x} - \mu 0)/(\sigma/\sqrt{n})$
Critical value −z(α)	-2.3263	NORMSINV(α)
p-value (left tail)	0.00172	Φ(z)
(c) Decision (critical value method)	**Reject H0**	
(c) Decision (p-value method)	**Reject H0**	

Conclusion: At α = 0.01, mean price in the South is significantly less than the national mean.

Ex.3 Breaking strength of ribbon – t-test (left-tailed)

H0: μ = 180 vs H1: μ < 180 (σ unknown, n small, population normal → t-test)

Hypothesised mean μ0 (pounds)	180	
Sample size n	5	
Sample mean x̄	169.5	
Sample SD s	5.7	
Significance level α	0.01	
Degrees of freedom	4	df = n − 1
Test statistic t	-4.1191	$t = (\bar{x} - \mu 0)/(s/\sqrt{n})$
Critical value −t(α, df)	-3.7469	−TINV(2α, df)
p-value (left tail)	0.00731	TDIST(|t|, df, 1)
Decision (critical value method)	**Reject H0**	
Decision (p-value method)	**Reject H0**	

Conclusion: Mean breaking strength is significantly below 180 pounds.

Ex.4 Lead content in water specimens (μg/L) – t-test (right-tailed)

H0: μ = 2.25 vs H1: μ > 2.25 (σ unknown → t-test on sample data)

2.4	2.9	2.7
2.6	2.9	2.0
2.8	2.2	2.4
2.4	2.0	2.5

Hypothesised mean μ0	2.25	
Significance level α	0.025	
Sample size n	12	COUNT
Sample mean x̄	2.4833	AVERAGE
Sample SD s	0.3129	STDEV (n−1)
Degrees of freedom	11	df = n − 1
Test statistic t	2.5836	$t = (\bar{x} - \mu 0)/(s/\sqrt{n})$
Critical value t(α, df)	2.2010	TINV(2α, df)
p-value (right tail)	0.01271	TDIST(t, df, 1)
Decision (critical value method)	**Reject H0**	
Decision (p-value method)	**Reject H0**	

Conclusion: Mean lead content is significantly greater than 2.25 μg/L.